Wide odd-field root validation and direct arithmetic checkpoint

Date: 2026-10-17T03:28:03.846728+00:00

All66 known-answer supplied quintics completed across11 odd fields: Fp3/5/7/11, P192/P224/P256/P384/P521, GF9/GF25. All field-root radicals, factor products, exact multiplicities and interval root sets matched planted factors or a certified irreducible-quadratic witness. This is a classical univariate component validation, not a generalized hard-system solve, fastest-external comparison or novelty claim.

The wide worker includes canonical all-field roots outside the interval for certificate replay. Nonrational repeated factors are not converted to field roots. Capped or late computations would be inconclusive with null exact counts; no such outcome was accepted as empty.

Direct support multiplication: preregistered fixed cutoffs meeting AND<=0.75reference and standard clauses<=1.10reference on every panel: ['8']. Deterministic circuit counts omit common norm generation and all-root solving from both arms; they are not runtime or end-to-end gains.

Cutoff8: 83 AND=0.6116,clauses=0.8168, 163 AND=0.4256,clauses=0.6046, 233 AND=0.4245,clauses=0.5540, 283 AND=0.3579,clauses=0.5179, 409 AND=0.3371,clauses=0.4563, 571 AND=0.2680,clauses=0.3969.

Cutoff16: 83 AND=0.7571,clauses=0.8559, 163 AND=0.5665,clauses=0.6672, 233 AND=0.5653,clauses=0.6376, 283 AND=0.4369,clauses=0.5409, 409 AND=0.4344,clauses=0.5073, 571 AND=0.3269,clauses=0.4141.

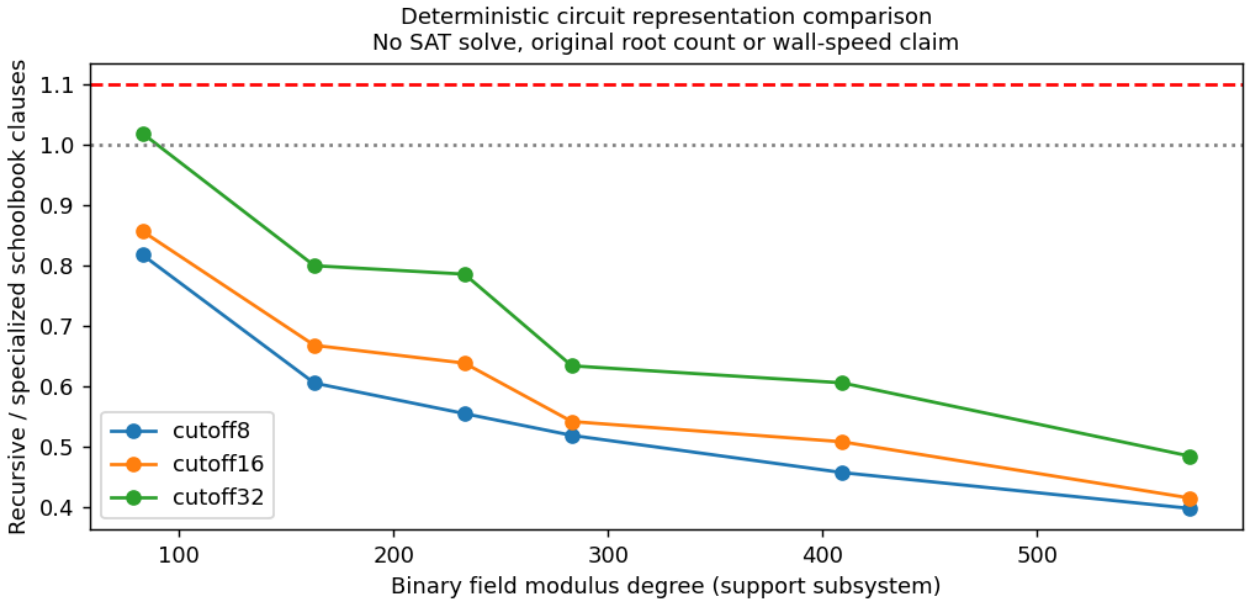
Cutoff32: 83 AND=1.0000,clauses=1.0182, 163 AND=0.7532,clauses=0.7992, 233 AND=0.7523,clauses=0.7851, 283 AND=0.5754,clauses=0.6332, 409 AND=0.5641,clauses=0.6050, 571 AND=0.4309,clauses=0.4836.

Native controls:3624 independent small odd quintics with Hasse multiplicities;27648 direct coefficient assignments/mutations and288 separate adaptive convolution vectors. New source and integration tests are in PR2049; CI is not represented as passed.

The original339-variable target, complete outer/squarefree search, source permutations and all auxiliary fibers remain open. No qualified8x all-roots F4/F5/F6 comparison or priority result is established. Prime-width univariate roots do not establish index-calculus relation search at those widths.

Next: independent strong external field-root crosscheck, integration of the selected arithmetic into a complete original-equivalent solver, and complementary squarefree/global norm search. Preserve every failed/inconclusive arm and all fixed cutoff regressions.

Reports remain local because automatic approval review rejected benchmark publication without explicit payload authorization. No alternate upload format bypasses that rejection.


Deterministic circuit representation comparison
No SAT solve, original root count or wall-speed claim
Recursive / specialized schoolbook clauses
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
cutoff8
cutoff16
cutoff32
100
200
300
400
500
Binary field modulus degree (support subsystem)